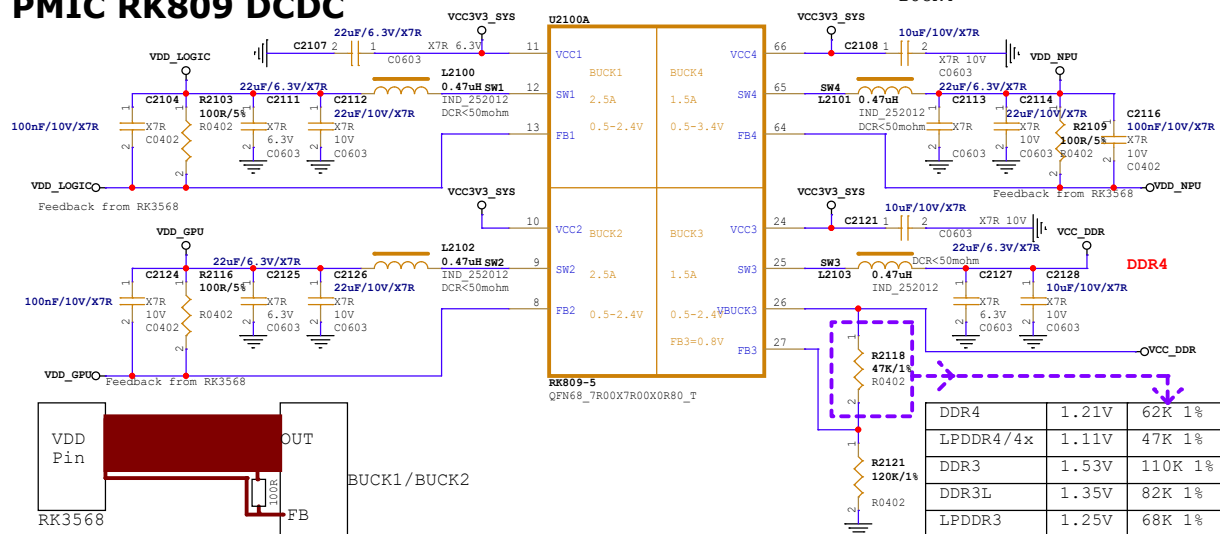
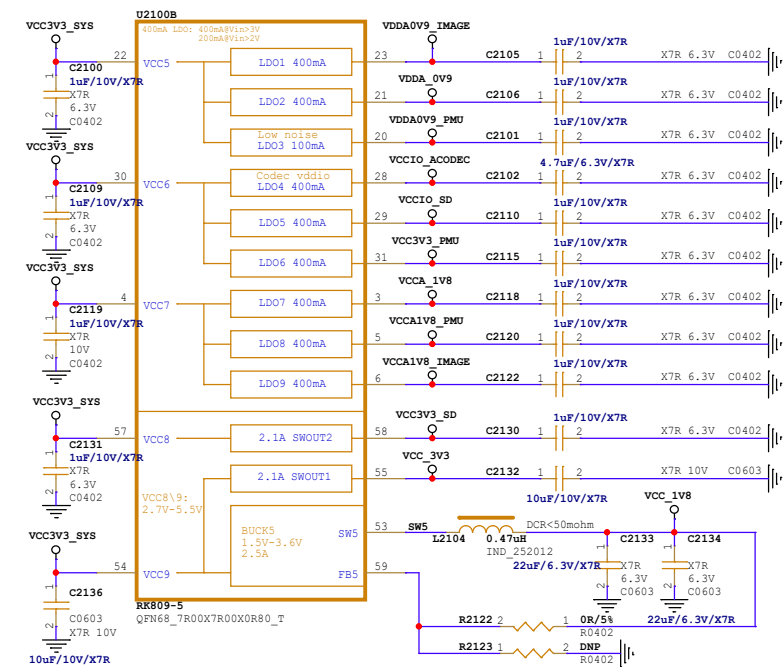


PMIC RK809 DCDC

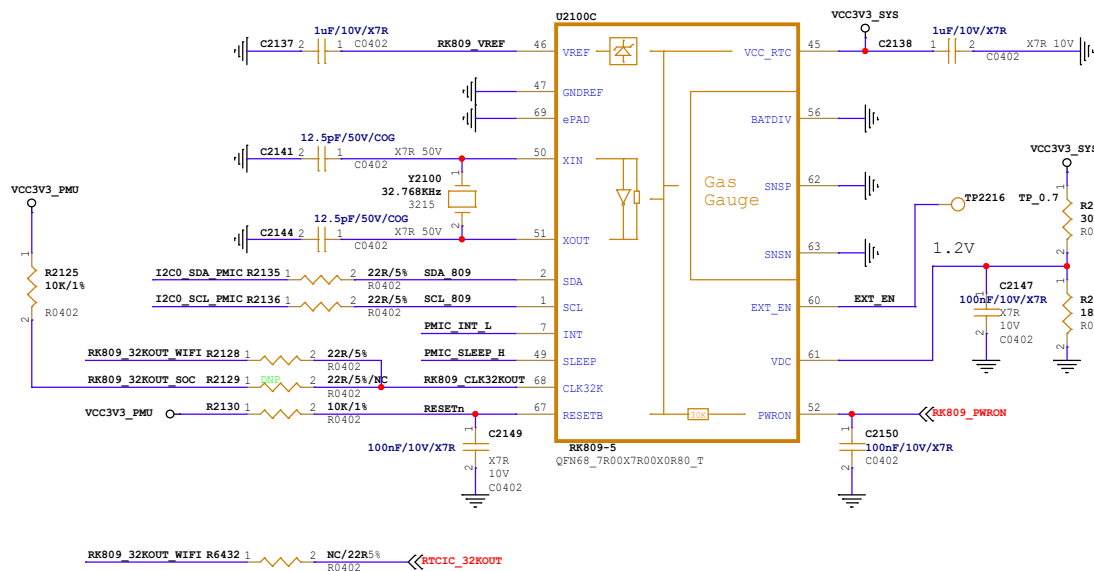


DDR4	1.21V	62K 1%
LPDDR4/4x	1.11V	47K 1%
DDR3	1.53V	110K 1%
DDR3L	1.35V	82K 1%
LPDDR3	1.25V	68K 1%

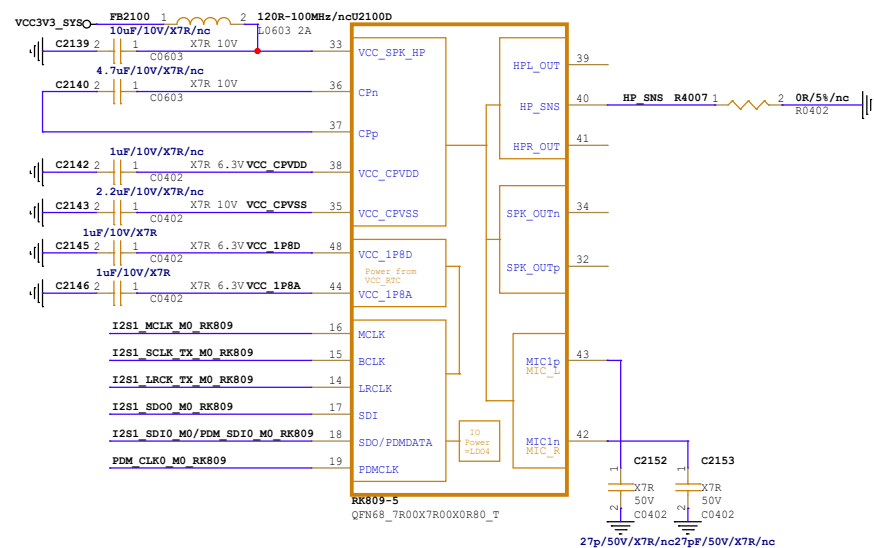
PMIC RK809 LDO



PMIC RK809 Managerment



PMIC RK809 CODEC



RK3568_H (VCCIO1 Domain)

U1000H

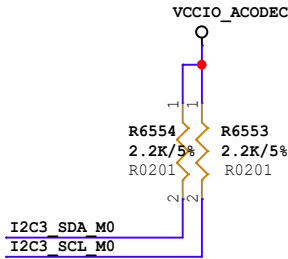
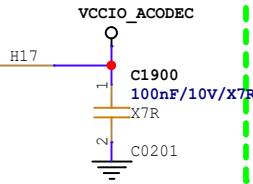
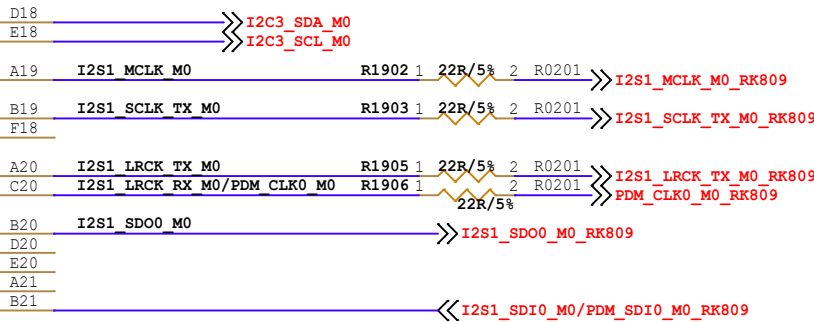
VCCIO1 Domain

Operating Voltage=1.8V/3.3V

I2C3 SDA M0	/ UART3 RX M0	/ CAN1 RX M0	/ AUDIOPWM LOUT P	/ ACODEC ADC DATA	/ GPIO1 A0 u
I2C3 SCL M0	/ UART3 TX M0	/ CAN1 TX M0	/ AUDIOPWM LOUT N	/ ACODEC ADC CLK	/ GPIO1 A1 u
I2S1 MCLK M0	/ UART3 RTSn M0	/ SCR CLK	/ PCIE30X1 PERSTn M2		/ GPIO1 A2 d
I2S1 SCLK TX M0	/ UART3 CTSn M0	/ SCR IO	/ PCIE30X1 WAKEn M2	/ ACODEC DAC CLK	/ GPIO1 A3 d
I2S1 SCLK RX M0	/ UART4 RX M0	/ PDM CLK1 M0	/ SPDIF TX M0		/ GPIO1 A4 d
I2S1 LRCK TX M0	/ UART4 RTSn M0	/ SCR RST	/ PCIE30X1 CLKREOn M2	/ ACODEC DAC SYNC	/ GPIO1 A5 d
I2S1 LRCK RX M0	/ UART4 TX M0	/ PDM CLK0 M0	/ AUDIOPWM ROUT P		/ GPIO1 A6 d
I2S1 SDO0 M0	/ UART4 CTSn M0	/ SCR DET	/ AUDIOPWM ROUT N	/ ACODEC DAC DATAL	/ GPIO1 A7 d
I2S1 SDO1 M0	/ I2S1 SDI3 M0	/ PDM SDI3 M0	/ PCIE20 CLKREOn M2	/ ACODEC DAC DATAR	/ GPIO1 B0 d
I2S1 SDO2 M0	/ I2S1 SDI2 M0	/ PDM SDI2 M0	/ PCIE20 WAKEn M2	/ ACODEC ADC SYNC	/ GPIO1 B1 d
I2S1 SDO3 M0	/ I2S1 SDI1 M0	/ PDM SDI1 M0	/ PCIE20 PERSTn M2		/ GPIO1 B2 d
	/ I2S1 SDI0 M0	/ PDM SDI0 M0			/ GPIO1 B3 d

RK3568-Socket
BGA636_RK3568

Default:RK809+PDM MIC
S1900=ON
S1901=OFF



Note:

Caps of between dashed green lines and U1000 should be placed under the U1000 package

Xtremer Shenzhen zhuodian Electronic Technology Co., Ltd.			
Project:	KP-06S_RK3568		
File:	11.RK3568_Audio Interface		
Date:	Tuesday, September 10, 2024	Rev:	<Revision>
Designed by:	<designer>	Reviewed by:	<Checker>
Sheet:	0	of	0

RK3568_L (VCCIO5 Domain)

U1000L

VCCIO5 Domain

Operating Voltage=1.8V/3.3V

LCDC D0	/ VOP BT656 D0 M0	/ SPI0 MISO M1	/ PCIE20 CLKREQn M1	/ I2S1 MCLK M2	/ GPIO2 D0 d
LCDC D1	/ VOP BT656 D1 M0	/ SPI0 MOSI M1	/ PCIE20 WAKEn M1	/ I2S1 SCLK TX M2	/ GPIO2 D1 d
LCDC D2	/ VOP BT656 D2 M0	/ SPI0 CS0 M1	/ PCIE30X1 CLKREQn M1	/ I2S1 LRCK TX M2	/ GPIO2 D2 d
LCDC D3	/ VOP BT656 D3 M0	/ SPI0 CLK M1	/ PCIE30X1 WAKEn M1	/ I2S1 SDIO M2	/ GPIO2 D3 d
LCDC D4	/ VOP BT656 D4 M0	/ SPI2 CS1 M1	/ PCIE30X2 CLKREQn M1	/ I2S1 SDI1 M2	/ GPIO2 D4 d
LCDC D5	/ VOP BT656 D5 M0	/ SPI2 CS0 M1	/ PCIE30X2 WAKEn M1	/ I2S1 SDI2 M2	/ GPIO2 D5 d
LCDC D6	/ VOP BT656 D6 M0	/ SPI2 MOSI M1	/ PCIE30X2 PERSTn M1	/ I2S1 SDI3 M2	/ GPIO2 D6 d
LCDC D7	/ VOP BT656 D7 M0	/ SPI2 MISO M1	/ UART8 TX M1	/ I2S1 SDO0 M2	/ GPIO2 D7 d
LCDC CLK	/ VOP BT656 CLK M0	/ SPI2 CLK M1	/ UART8 RX M1	/ I2S1 SDO1 M2	/ GPIO3 A0 d
LCDC D8	/ VOP BT1120 D0	/ SPI1 CS0 M1	/ PCIE30X1 PERSTn M1	/ SDMMC2 D0 M1	/ GPIO3 A1 d
LCDC D9	/ VOP BT1120 D1	/ GMAC1 TXD2 M0	/ I2S3 MCLK M0	/ SDMMC2 D1 M1	/ GPIO3 A2 d
LCDC D10	/ VOP BT1120 D2	/ GMAC1 TXD3 M0	/ I2S3 SCLK M0	/ SDMMC2 D2 M1	/ GPIO3 A3 d
LCDC D11	/ VOP BT1120 D3	/ GMAC1 RXD2 M0	/ I2S3 LRCK M0	/ SDMMC2 D3 M1	/ GPIO3 A4 d
LCDC D12	/ VOP BT1120 D4	/ GMAC1 RXD3 M0	/ I2S3 SDO M0	/ SDMMC2 CMD M1	/ GPIO3 A5 d
LCDC D13	/ VOP BT1120 CLK	/ GMAC1 TXCLK M0	/ I2S3 SDI M0	/ SDMMC2 CLK M1	/ GPIO3 A6 d
LCDC D14	/ VOP BT1120 D5	/ GMAC1 RXCLK M0	/ SDMMC2 DET M1	/ GPIO3 A7 d	
LCDC D15	/ VOP BT1120 D6	/ ETH1 REFCLK0 25M M0	/ SDMMC2 PWREN M1	/ GPIO3 B0 d	
LCDC D16	/ VOP BT1120 D7	/ GMAC1 RXD0 M0	/ UART4 RX M1	/ PWM8 M0	/ GPIO3 B1 d
LCDC D17	/ VOP BT1120 D8	/ GMAC1 RXD1 M0	/ UART4 TX M1	/ PWM9 M0	/ GPIO3 B2 d
LCDC D18	/ VOP BT1120 D9	/ GMAC1 RXDV CRS M0	/ I2C5 SCL M0	/ PDM SDIO M2	/ GPIO3 B3 d
LCDC D19	/ VOP BT1120 D10	/ GMAC1 RXER M0	/ I2C5 SDA M0	/ PDM SDI1 M2	/ GPIO3 B4 d
LCDC D20	/ VOP BT1120 D11	/ GMAC1 TXD0 M0	/ I2C3 SCL M1	/ PWM10 M0	/ GPIO3 B5 d
LCDC D21	/ VOP BT1120 D12	/ GMAC1 TXD1 M0	/ I2C3 SDA M1	/ PWM11 IR M0	/ GPIO3 B6 d
LCDC D22	/ PWM12 M0	/ GMAC1 TXEN M0	/ UART3 TX M1	/ PDM SDI2 M2	/ GPIO3 B7 d
LCDC D23	/ PWM13 M0	/ GMAC1 MCLKINOUT M0	/ UART3 RX M1	/ PDM SDI3 M2	/ GPIO3 C0 d
LCDC HSYNC	/ VOP BT1120 D13	/ SPI1 MOSI M1	/ PCIE20 PERSTn M1	/ I2S1 SDO2 M2	/ GPIO3 C1 d
LCDC VSYNC	/ VOP BT1120 D14	/ SPI1 MISO M1	/ UART5 TX M1	/ I2S1 SDO3 M2	/ GPIO3 C2 d
LCDC DEN	/ VOP BT1120 D15	/ SPI1 CLK M1	/ UART5 RX M1	/ I2S1 SCLK RX M2	/ GPIO3 C3 d
PWM14 M0	/ VOP PWM M1	/ GMAC1 MDC M0	/ UART7 TX M1	/ PDM CLK1 M2	/ GPIO3 C4 d
PWM15 IR M0	/ SPDIF TX M1	/ GMAC1 MDIO M0	/ UART7 RX M1	/ I2S1 LRCK RX M2	/ GPIO3 C5 d

RK3568-Socket
BGA636_RK3568

Note:

Caps of between dashed green lines and U1000 should be placed under the U1000 package

AG6
AD7
AC8
AC7
AF5
AF6
AD6
AH5
AH4
AB8
AE5
AG4
AF4
AH3
AG3
AH2
AG2
AG1
AF2
AF1
AE1
AE2
AE3
AD4
AD2
AD1
AA7
AC4
AC3
AC2

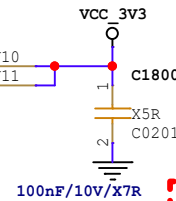
GPIO3 A1_d
GPIO3 A2_d
TYPEC0_INT

I2C5_SCL_M0
I2C5_SDA_M0

TP2214 TP_0.7
TP2215 TP_0.7

UART5_TX_M1
UART5_RX_M1

UART7_TX_M1
UART7_RX_M1



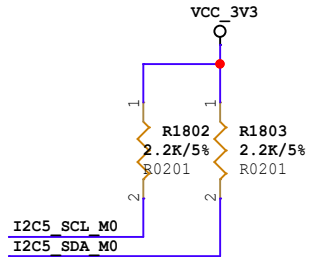
100nF/10V/x7R

Note:

If VCCIO5 domain power voltage is adjusted, the software DTS configuration must be updated synchronously, otherwise the IO may be damaged!

If the VCCIO5 hardware has been modified to 1.8V power supply, and the corresponding DTS must be modified to 1.8V configuration, otherwise the IO function of VCCIO5 will be abnormally.

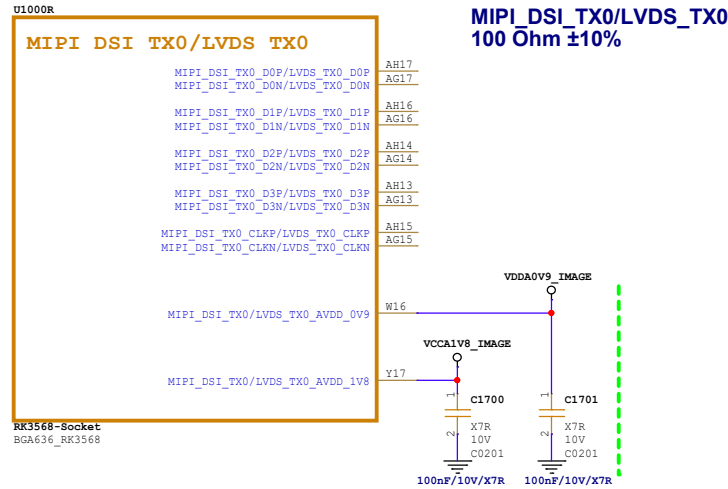
The VCCIO5 hardware has been modified to 3.3V power supply, if the software DTS configuration is still 1.8V configuration, the IO of VCCIO5 will be damaged!



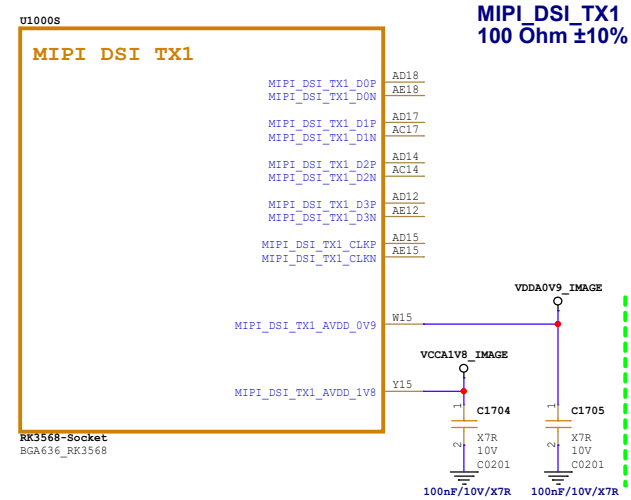
for PCIE WIFI

Xtremer® Shenzhen zhuodian Electronic Technology Co., Ltd.			
Project:	KP-06S_RK3568		
File:	12.RK3568_VO Interface_2		
Date:	Wednesday, September 25, 2024	Rev:	<Revision>
Designed by:	<designer>	Reviewed by:	<Checker>
Sheet:	0	of	0

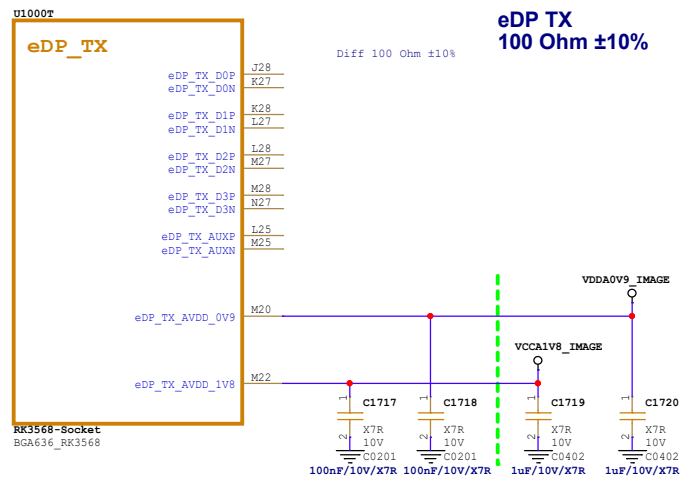
RK3568_R (MIPI_DSI_TX0/LVDS_TX0)



RK3568_S (MIPI_DSI_TX1)



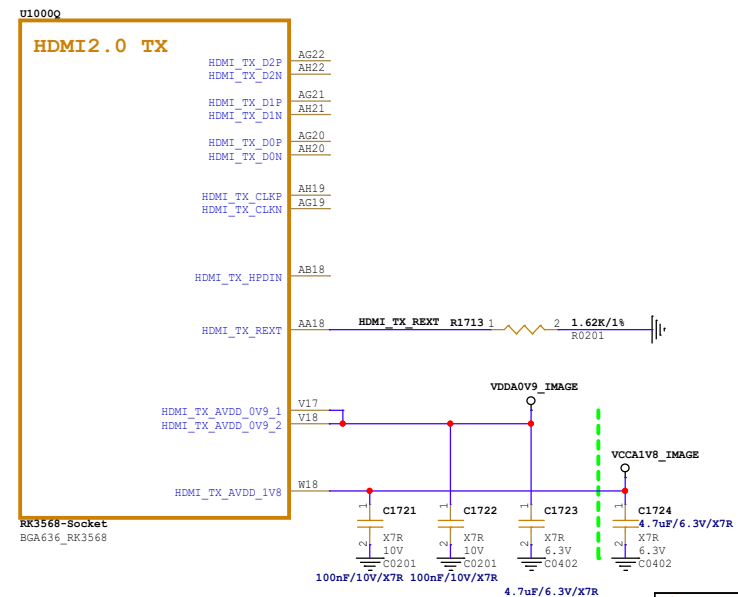
RK3568_T (eDP TX)



Note:

Caps of between dashed green lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

RK3568_Q (HDMI2.0 TX)

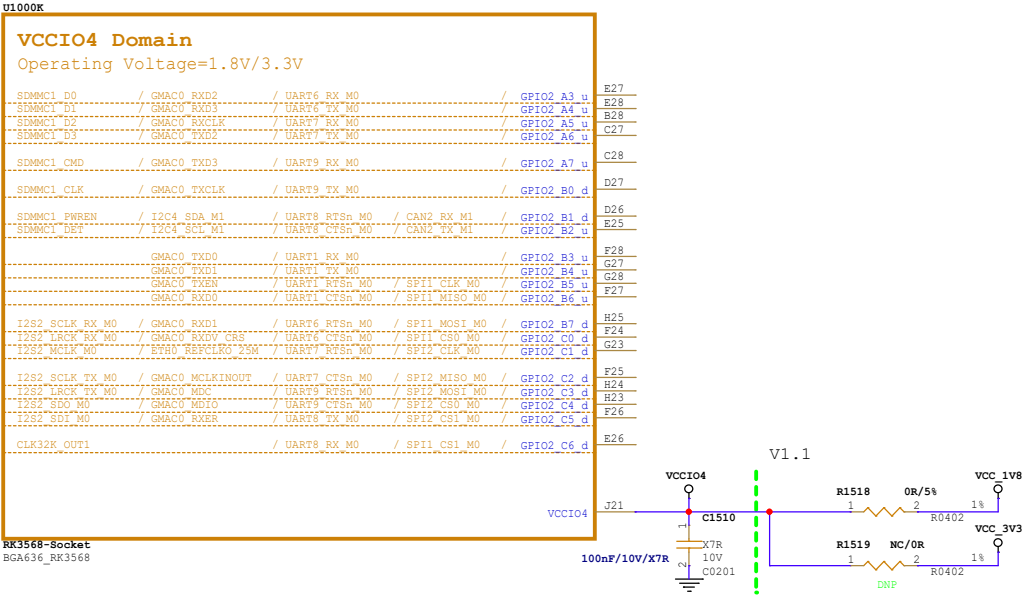


[illegible][illegible]

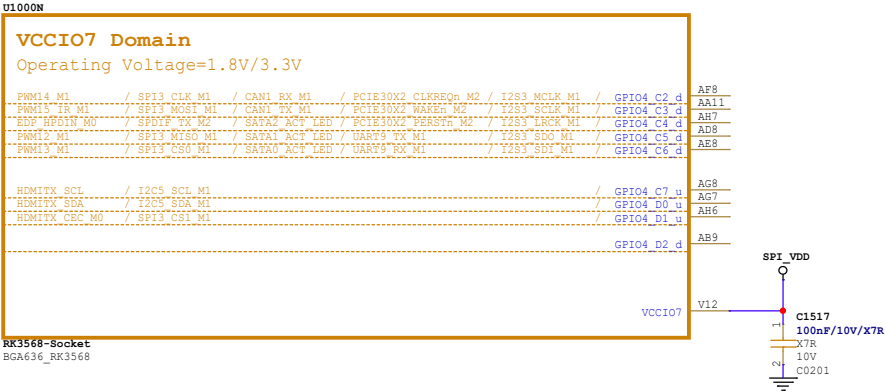
```
Support BT601 YCbCr 422 8bit input
Support BT656 YCbCr 422 8bit input
Support RAW 8/10/12bit input
Support BT1120 YCbCr 422 8/10/12/16bit input, single/dual-edge sampling
Support 2/4 mixed BT656/BT1120 YCbCr 422 8bit input
```

 Shenzhen zhuodian Electronic Technology Co., Ltd.			
Project:		KP-06S_RK3568	
File:		06.RK3568_VI interface	
Date:	Friday, September 27, 2024	Rev:	<Revision>
Designed by:	<designer>	Reviewed by:	<checker>
		Sheet:	0 of 0

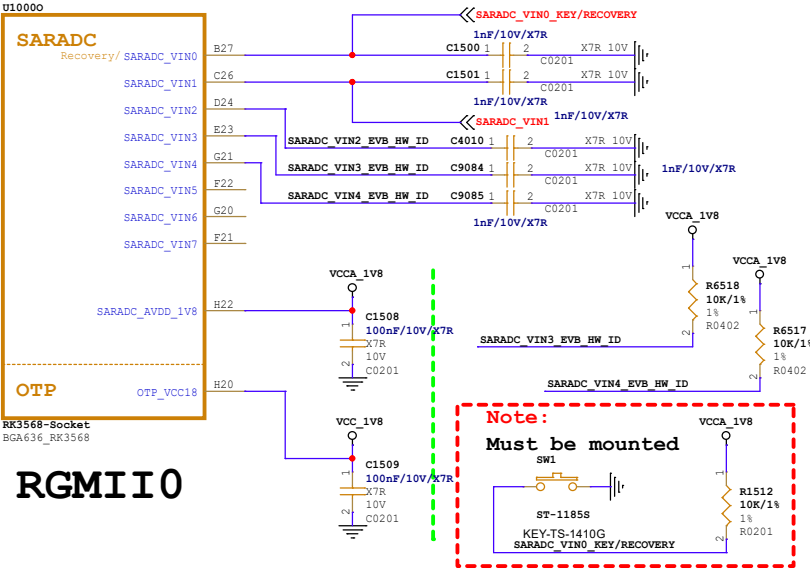
RK3568_K (VCCIO4 Domain)



RK3568_N (VCCIO7 Domain)

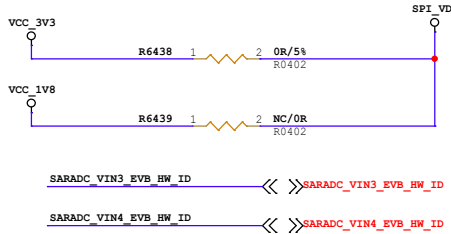


RK3568_O (SARADC/OTP)

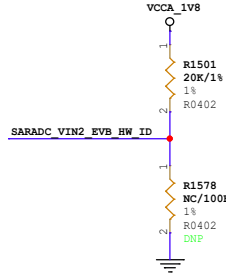


RGMIIO

SPI VDD

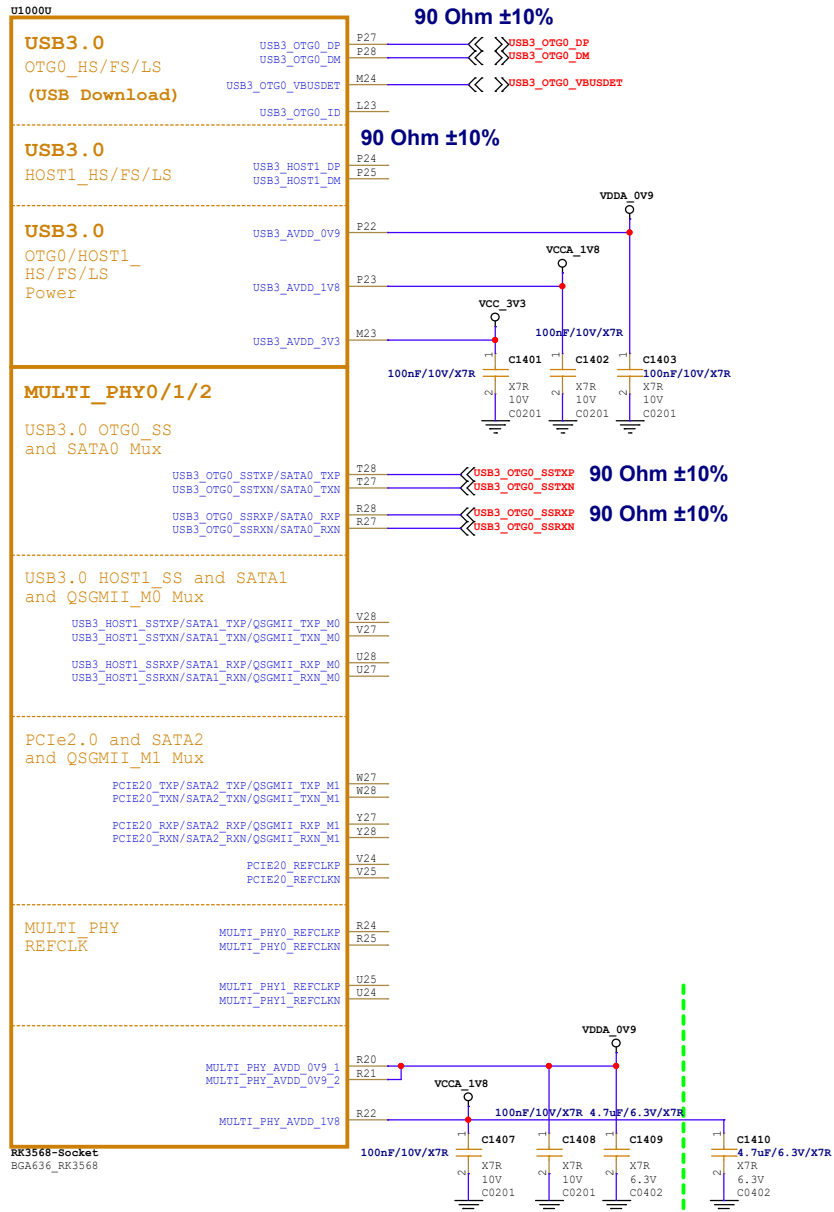


SARADC_VIN1_EVB_HW_ID	Rup	Rdown	ADC	
EVB1	10K	DNP	1023	1.8V
EVB2	20K	100K	852	1.5V
EVB3	18K	36K	681	1.2V
EVB4	51K	51K	512	0.9V
EVB5	36K	18K	340	0.6V
EVB6	100K	20K	170	0.3V
EVB7	DNP	10K	0	0V
EVB8				



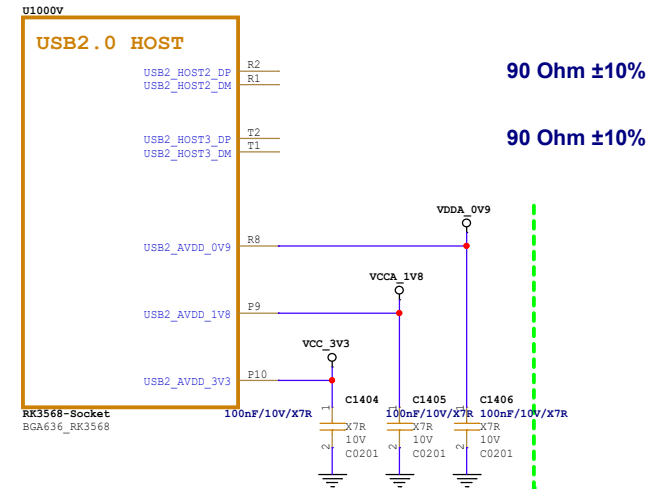
Note:
Caps of between dashed green lines and U1000 should be placed under the U1000 package

RK3568 V (USB2.0 HOST)

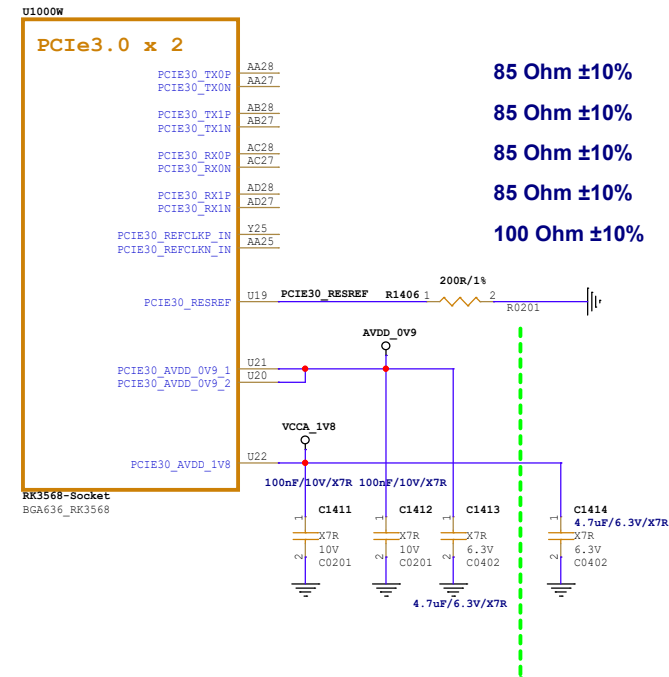


Note:

Caps of between dashed green lines and U1000 should be placed under the U1000 package. Other caps should be placed close to the U1000 package

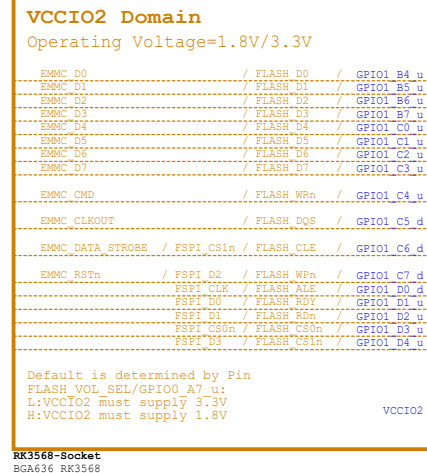


RK3568_W (PCIe3.0 x2)



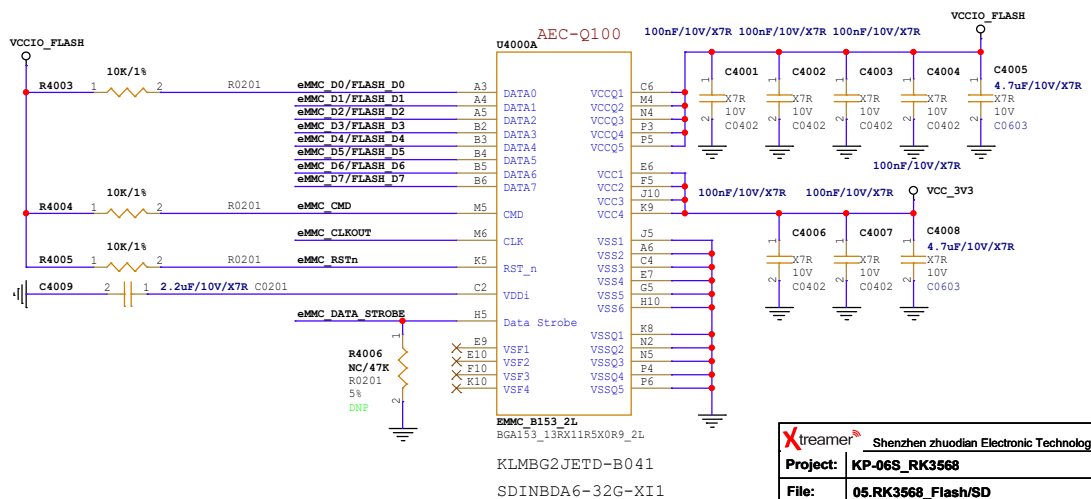
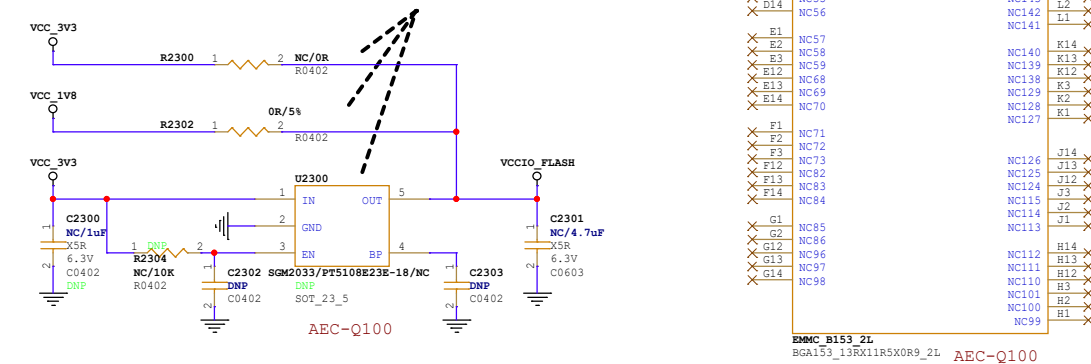
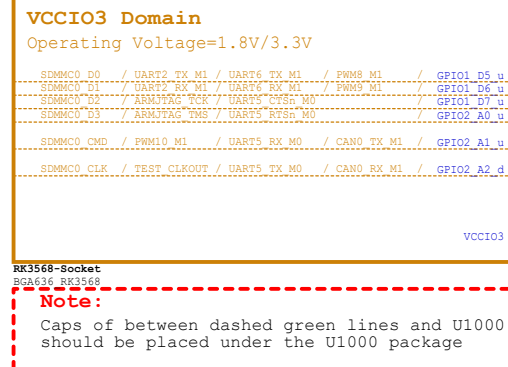
RK3568_I (VCCIO2 Domain)

U1000I

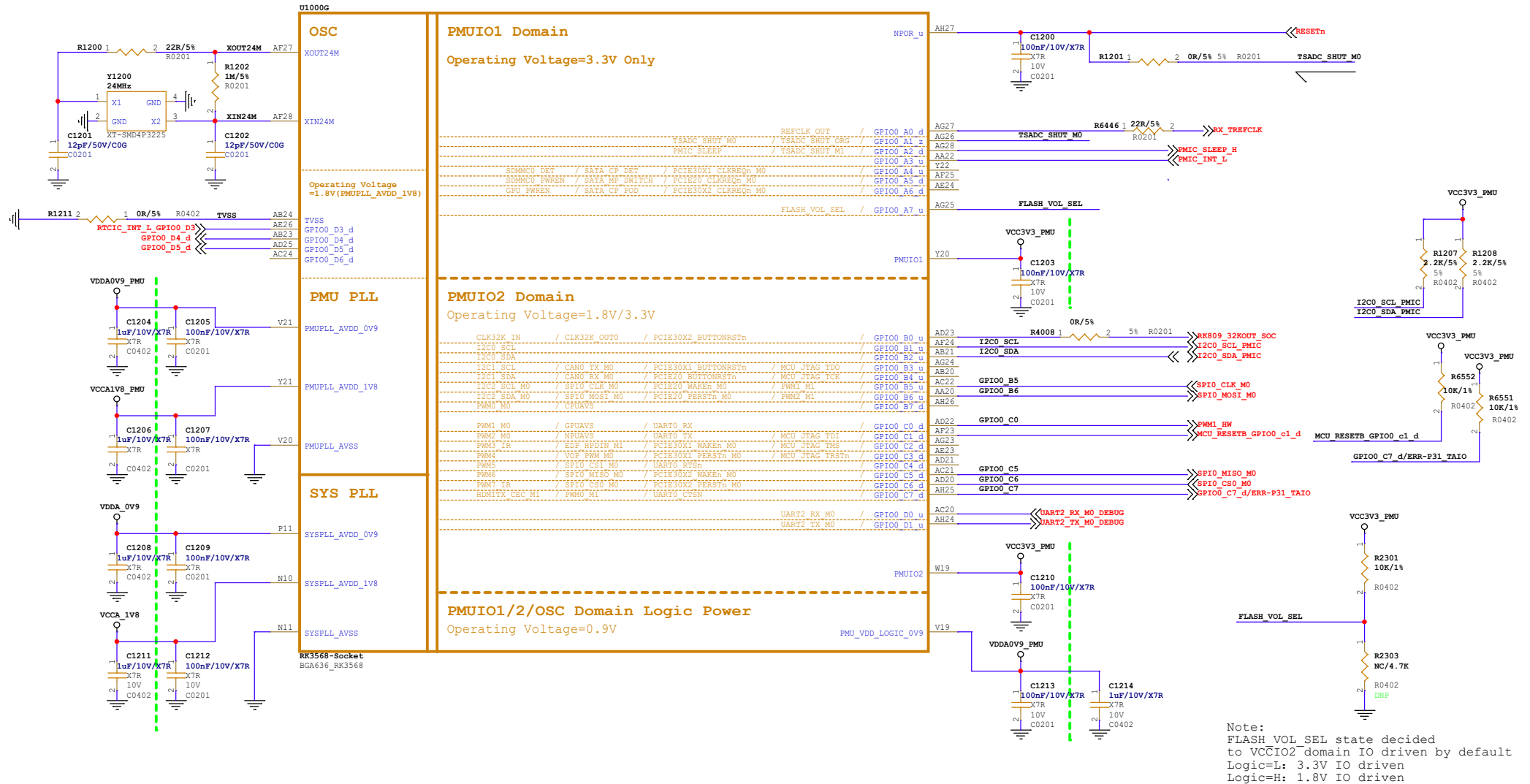


RK3568_J (VCCIO3 Domain)

U1000J



RK3568_G (OSC/PLL/PMUIO1/2)

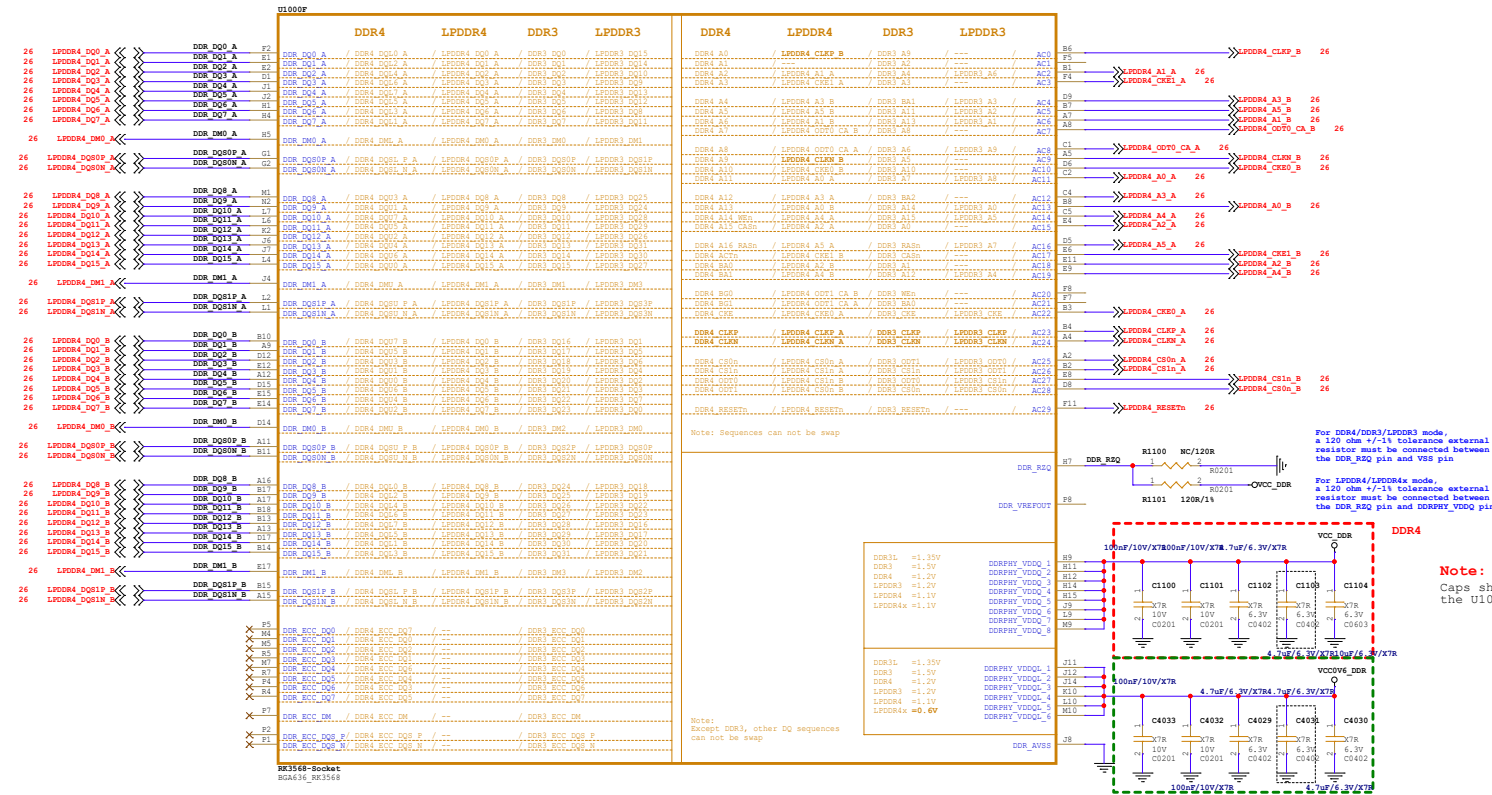


Note:

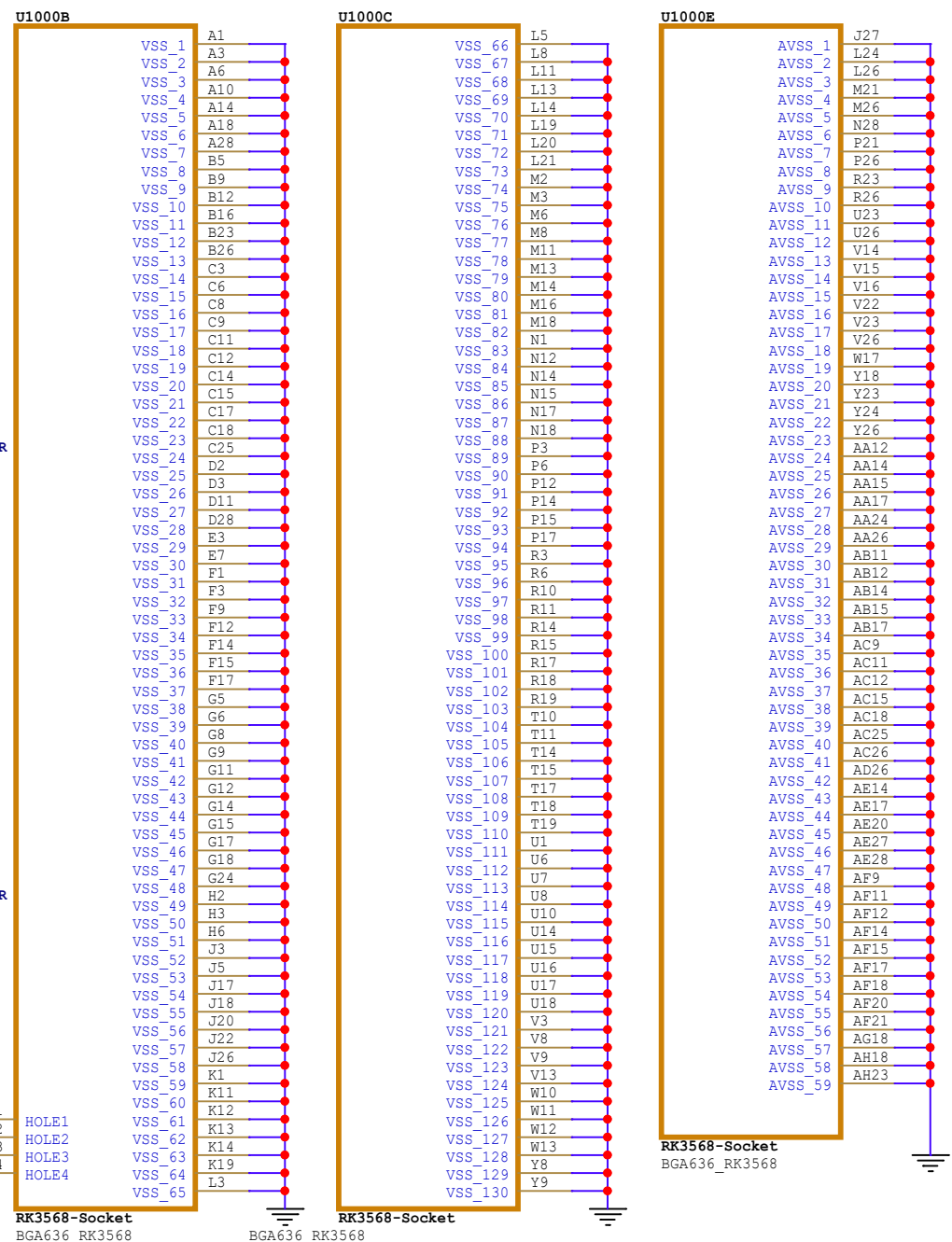
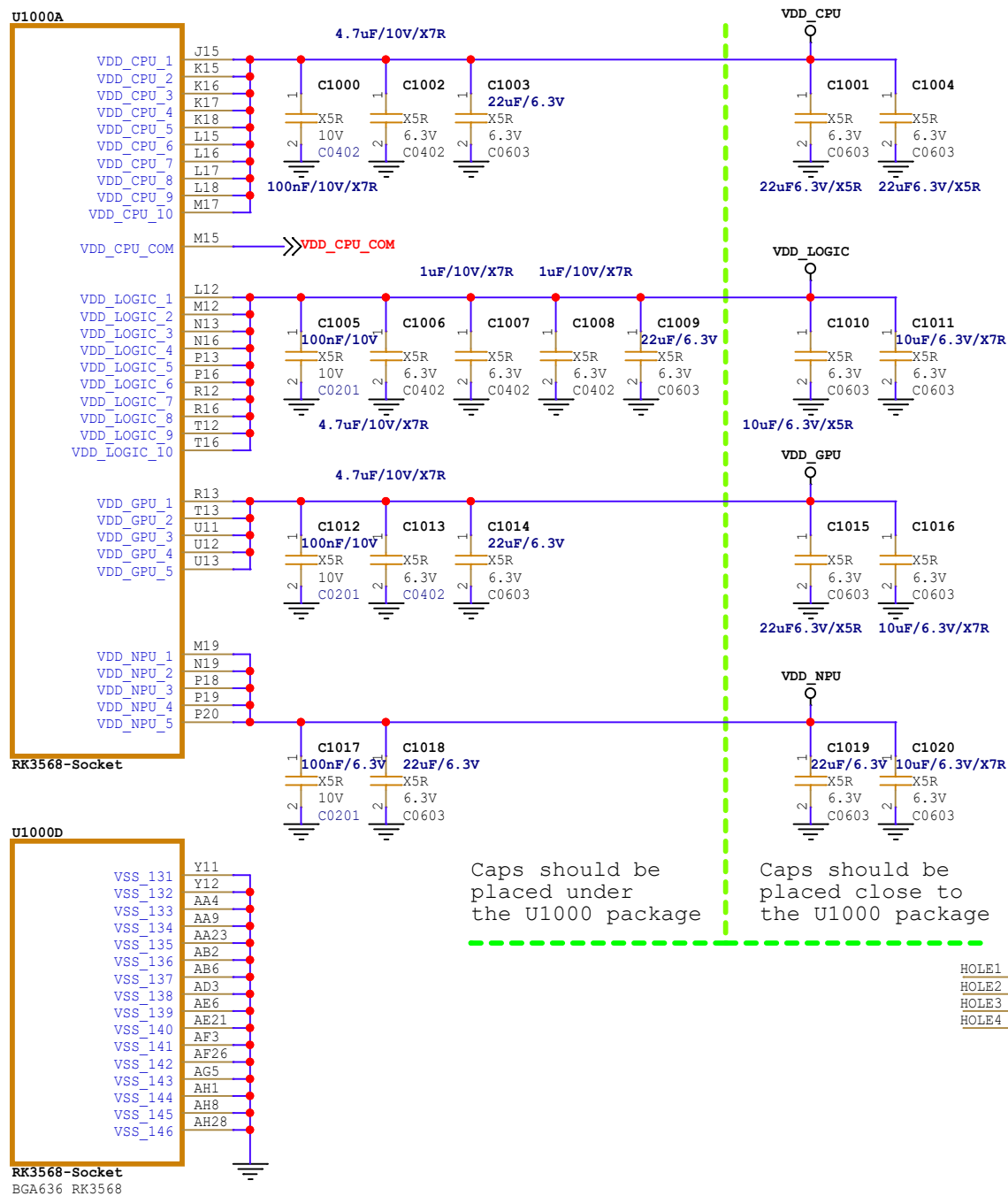
Caps of between dashed green lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

Note:
FLASH VOL SEL state decided to VCCIO2 domain IO driven by default
Logic=L: 3.3V IO driven
Logic=H: 1.8V IO driven

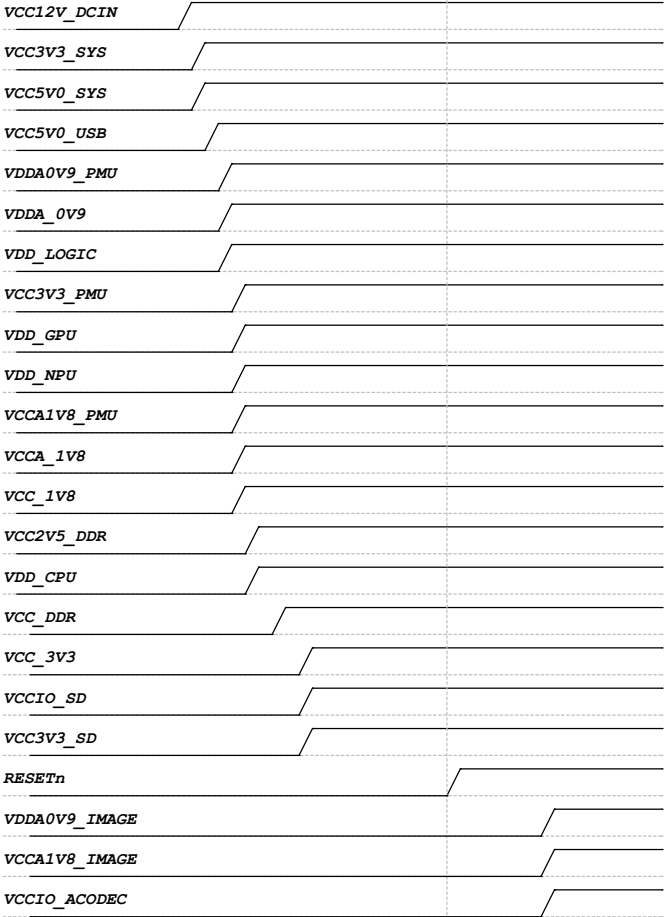
RK3568_F (DDR PHY)



RK3568 ABCDE (Power&Gnd)



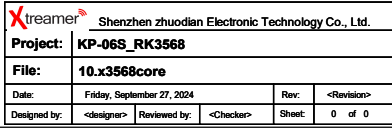
Power Sequence

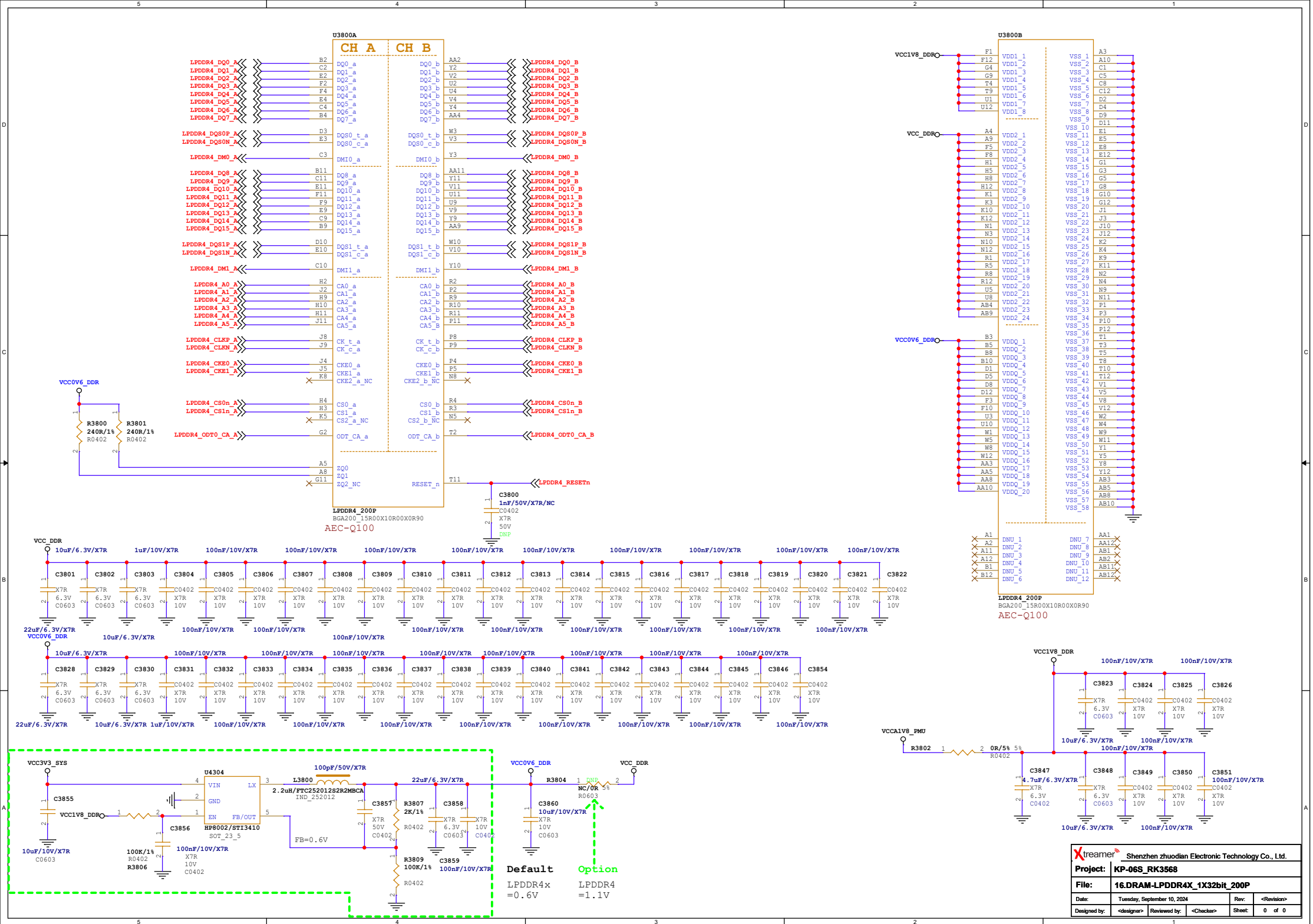


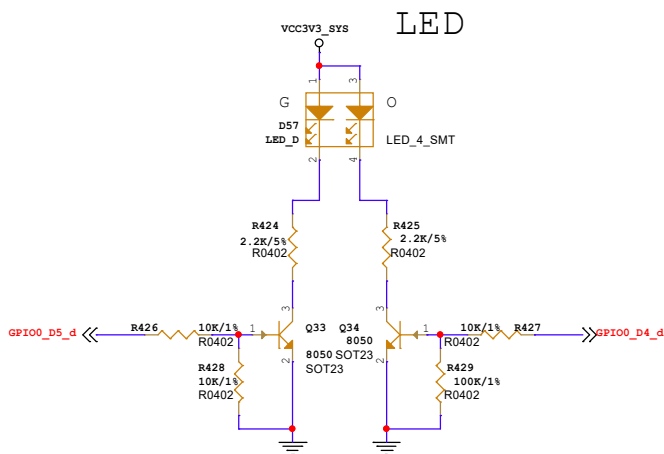
Power Supply	PMIC Channel	Supply Limit	Power Name	Time Slot	Default Voltage	Default ON/OFF	Sleep ON/OFF	Peak Current	Sleep Current
VCC3V3_SYS	RK809_BUCK1	2.5A	VDD_LOGIC	Slot:1	0.9V	ON	OFF	TBD	TBD
VCC3V3_SYS	RK809_BUCK2	2.5A	VDD_GPU	Slot:2	0.9V	ON	OFF	TBD	TBD
VCC3V3_SYS	RK809_BUCK3	1.5A	VCC_DDR	Slot:3	ADJ FB=0.8V	ON	ON	TBD	TBD
VCC3V3_SYS	RK809_BUCK4	1.5A	VDD_NPU	N/A	0.9V	OFF	OFF	TBD	TBD
VCC3V3_SYS	RK809_LDO1	0.4A	VDDA0V9_IMAGE	N/A	0.9V	OFF	OFF	TBD	TBD
	RK809_LDO2	0.4A	VDDA_0V9	Slot:1	0.9V	ON	OFF	TBD	TBD
	RK809_LDO3	0.1A	VDDA0V9_PMU	Slot:1	0.9V	ON	ON	TBD	TBD
VCC3V3_SYS	RK809_LDO4	0.4A	VCCIO_ACODEC	N/A	3.3V	OFF	OFF	TBD	TBD
	RK809_LDO5	0.4A	VCCIO_SD	Slot:4	3.3V	ON	OFF	TBD	TBD
	RK809_LDO6	0.4A	VCC3V3_PMU	Slot:2	3.3V	ON	ON	TBD	TBD
VCC3V3_SYS	RK809_LDO7	0.4A	VCCA_1V8	Slot:2	1.8V	ON	OFF	TBD	TBD
	RK809_LDO8	0.4A	VCCA1V8_PMU	Slot:2	1.8V	ON	ON	TBD	TBD
	RK809_LDO9	0.4A	VCCA1V8_IMAGE	N/A	1.8V	OFF	OFF	TBD	TBD
VCC3V3_SYS	RK809_SW2 100mohm	2.1A	VCC3V3_SD	Slot:4	3.3V	ON	OFF	TBD	TBD
VCC3V3_SYS	RK809_SW1 90mohm	2.1A	VCC_3V3	Slot:4	3.3V	ON	OFF	TBD	TBD
	RK809_BUCK5	2.5A	VCC_1V8	Slot:2	1.8V	ON	OFF	TBD	TBD
	RK809_RESETh			Slot:4+5					
VCC12V_DCIN	EXT BUCK	3.0A	VCC3V3_SYS	Slot:0	3.3V	ON	ON	TBD	TBD
VCC12V_DCIN	EXT BUCK	3.0A	VCC5V0_SYS	Slot:0	5.0V	ON	OFF	TBD	TBD
VCC5V0_SYS	EXT BUCK	6.0A	VDD_CPU	Slot:2A	1.025V	ON	OFF	TBD	TBD
VCC3V3_SYS	EXT LDO	0.3A	VCC2V5_DDR	Slot:2A	2.5V	ON	ON	TBD	TBD

IO Power Domain Map
Updates must be Revision accordingly!

IO Domain	Pin Num	Support IO Voltage		Actual assigned IO Domain Voltage			Notes
		3.3V	1.8V	Supply Power Net Name	Power Source	Voltage	
PMUIO1	Pin Y20	✓	✗	VCC3V3_PMU	VCC3V3_PMU	3.3V	
PMUIO2	Pin W19	✓	✓	VCC3V3_PMU	VCC3V3_PMU	3.3V	
VCCIO1	Pin H17	✓	✓	VCCIO_ACODEC	VCCIO_ACODEC	3.3V	
VCCIO2	Pin H18	✓	✓	VCCIO_FLASH	VCC_1V8	1.8V	PIN "FLASH_VOL_SEL" must be logic High if VCCIO_FLASH=3.3V,FLASH_VOL_SEL must be logic low
VCCIO3	Pin L22	✓	✓	VCCIO_SD	VCCIO_SD	3.3V	
VCCIO4	Pin J21	✓	✓	VCCIO4	VCC_1V8	1.8V	
VCCIO5	Pin V10 Pin V11	✓	✓	VCCIO5	VCC_3V3	3.3V	
VCCIO6	Pin R9 Pin U9	✓	✓	VCCIO6	VCC_1V8	1.8V	
VCCIO7	Pin V12	✓	✓	VCCIO7	VCC_3V3	3.3V	

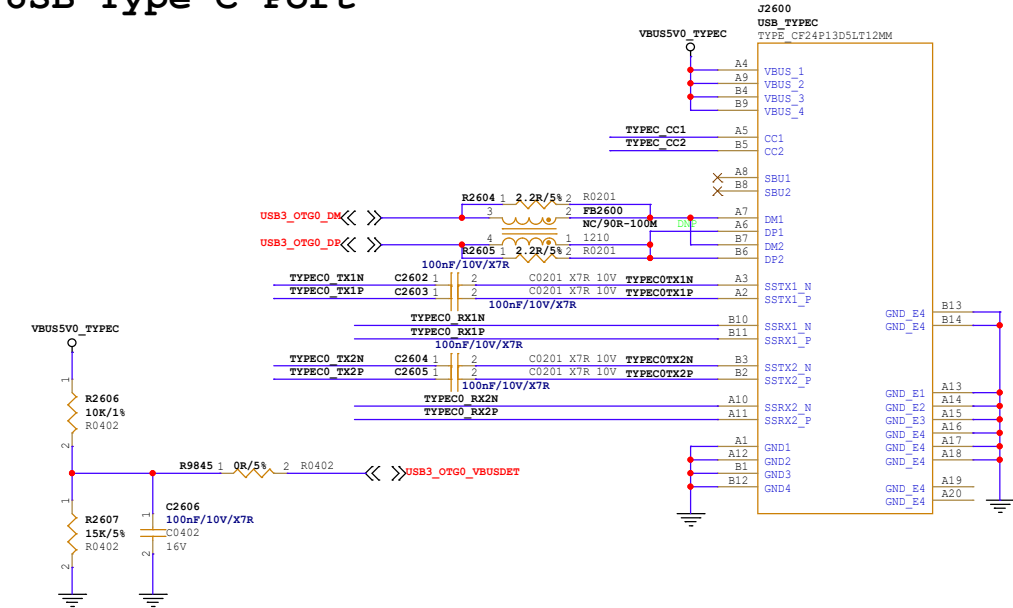




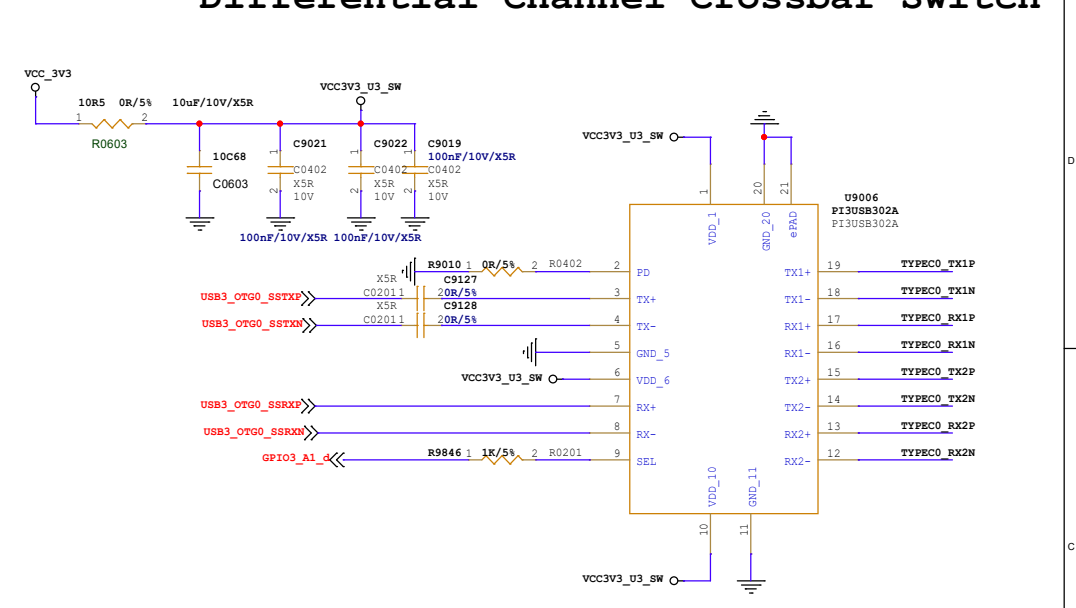


Xtremer [®] Shenzhen zhuodian Electronic Technology Co., Ltd.				
Project:	KP-06S_RK3568			
File:	17,LED			
Date:	Tuesday, September 10, 2024	Rev:	<Revision>	
Designed by:	<designer>	Reviewed by:	<Checker>	Sheet: 0 of 0

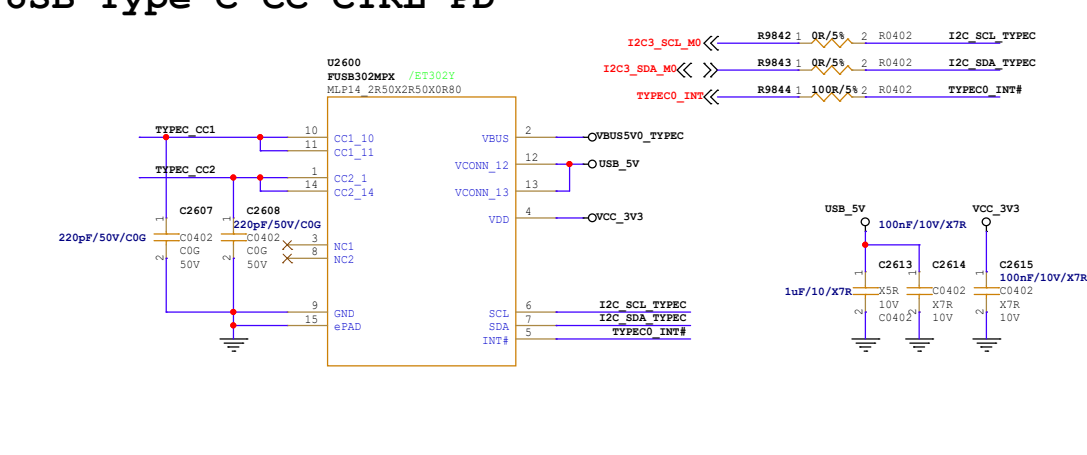
USB Type-C Port



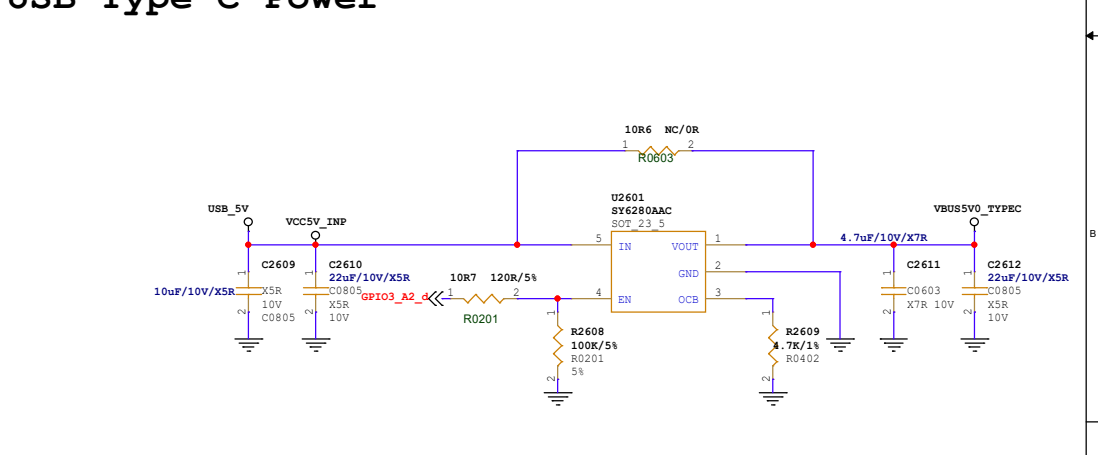
Differential Channel Crossbar Switch



USB Type-C CC CTRL PD

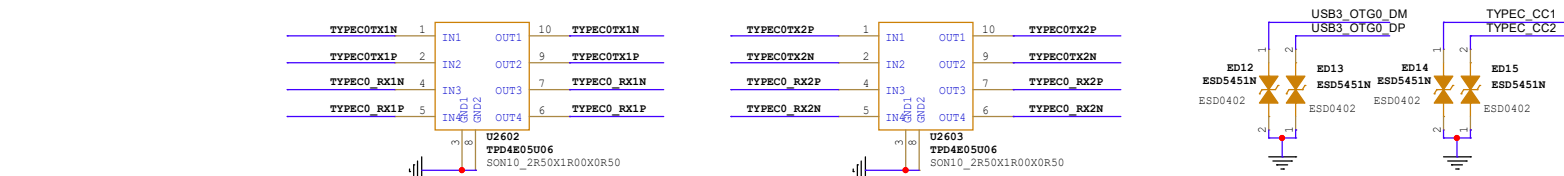


USB Type-C Power



ESD

Note:All the ESD components should be placed close to the port and Cj<=0.4pF



Xtremer® Shenzhen zhudian Electronic Technology Co., Ltd.			
Project:	KP-06S_RK3568		
File:	18.USB Type-C OTG		
Date:	Tuesday, September 10, 2024	Rev:	<Revision>
Designed by:	<designer>	Reviewed by:	<Checker>
Sheet:	0	of	0